Eddie Tang

609-480-1029 | eddietang2314@gmail.com | linkedin.com/in/ejtang | Portfolio | Skillman, NJ

Hardware-focused electrical engineering student with 3+ years of hands-on PCB and embedded firmware experience for 3V-24V avionics. Proven leader (team lead, VP) who drives cross-functional projects from prototype to product with strong bench validation and simulation skills.

Education

University of Illinois Urbana-Champaign — Urbana, IL
*BS in Electrical Engineering | GPA: **4.0** | James Scholar Honors* — *Aug. 2025 – May 2028*
Minor: Computer Science & Semiconductor Engineering

Montgomery High School (MHS) — Skillman, NJ
*High School Diploma | GPA: **3.92** | SAT: 1560 | 12 AP exams (all scores 5)* — *Sept. 2021 – Jun. 2025*

Relevant Coursework

Analog Signal Processing, Intro to Computing, Differential Equations, Multivariable Calculus, Quantum Physics

Projects

CAM-MK3 | Illinois Space Society - Spaceshot Avionics | KiCAD, C++, ESP32-IDF — Oct. 2025 – Present

- Spearheaded 3rd major revision of a compact live-video transmitter PCB for rocket avionics; co-designed schematic, layout, and power architecture to meet size, thermal, and RF constraints; established version control using Github
- Built and debugged multiple prototypes (reflow + hand-solder) and validated signal & power integrity with oscilloscope, digital scope probes, and bench power supplies.
- Authored and validated C++ I^2C and SPI firmware drivers for TVP5151 analog decoder, ESP32-P4 LCD_CAM interface, and NTSC→digital pipeline producing 720×480 color frames (YUV422/YUV420).
- Integrated on-device H.264 encoding and implemented UVC USB video streaming - enabling low-latency telemetry
- Implemented cross-board I^2C control and 433 MHz wireless communication for camera toggling and live-video feed

Other PCBs | Illinois Space Society - Spaceshot Avionics | KiCAD, C++ — Oct. 2025 – Present

- Designed GATOR power-distribution and ground-station PCBs to manage 24V systems and drive Yagi antenna servos; improved tracker reliability and maintainability through modular connector and test-point placement.

TARC Avionics V1–V3 | EasyEDA, C++, Python — Jun. 2022 – Jun. 2025

- Designed, assembled, and programmed custom PCBs and a 6-DOF AHRS using BMI088 IMU and MS5607 barometer for active flight control; achieved a reproducible 80% mission success across iterative flights.
- Developed quaternion-based sensor fusion & algorithm to integrate acceleration for accurate altitude estimation.
- Built Python software-in-the-loop simulations to validate control logic using historical flight telemetry.
- Ranked top 50 nationally at TARC Nationals (1,000 teams); qualified twice for Nationals.

Experience & Leadership

Electric Sensors Project Lead | *Illini Solar Car* — Jan. 2026 – Present

- Lead an 8-person team to design a Tire Pressure Monitoring System (TPMS) for ∼80 psi solar car wheels.
- Researched implementation of low-power operations, system-design, and protocols for BLE, IMUs, and others.
- Coordinated schedules and deliverables across hardware, firmware, and test subteams to meet project milestones.

Research Assistant (Paid), Mechanical Engineering | *Rowan University* — Summer 2025

- Investigated effects of point defects in thermoelectric materials on lattice thermal conductivity using MD.
- First author on in-review manuscript investigating point defect effects on thermal conductivity in Si and Ge.

Laboratory Learning Program, Mechanical & Aerospace Department | *Princeton University* — Summer 2024

- Performed DFT calculations on WO_3 surfaces for plasma-assisted NH_3 synthesis; analyzed results with NumPy.
- Co-authored a peer-reviewed publication in ACS Energy Letters (Impact Factor: 19.3).

Vice President, Aerospace Club & Founder, Rocketry Team | *MHS* — Sept. 2023 – Jun. 2025

- Led weekly meetings with 30+ students to engineer rockets; raised $15K+ through STEM fundraiser initiatives.
- Increased member attendance and participation by 10% by fostering a close-knit, community-first team culture.

Skills

Hardware & Tools: PCB schematic/layout (KiCAD, Eagle), reflow & prototype assembly, bench testing equipment
Languages & Firmware: C++, Python, C, Bash, Dart, Go, Java
Software & Frameworks: MCU IDEs (PlatformIO, MCUxpresso), VASP, LAMMPS, ASE, Slurm, Git, Jupyter, OnShape, NX Siemens, Excel
Other: HAM Radio Technician (KE2CNQ); Fluent in English and Chinese; NMSQT Finalist